

ABIN JOSEPH

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AI/ML Engineer | Generative AI Engineer

AI/ML & Generative AI Engineer with 3.8+ years of experience building and deploying Machine Learning, Generative AI, RAG, Agentic AI, and Document Intelligence solutions. Hands-on experience with Python, LangChain, LangGraph, Azure OpenAI, AWS Bedrock, FAISS, vector search, semantic retrieval, LLM evaluation, observability, computer vision, and FastAPI. Experienced in developing end-to-end AI pipelines covering document ingestion, preprocessing, embeddings, retrieval, LLM orchestration, evaluation, monitoring, and production deployment.

WORK EXPERIENCE

ThinkPalm Technologies Pvt. Ltd.

09/2022 - 07/2026

Software Engineer (AI/ML Focus)

- Designed and deployed end-to-end RAG systems (Python, LangChain, FAISS, Azure OpenAI) for regulatory and maritime compliance documents, replacing manual document search with instant, context-grounded answers.
- Built multi-agent LangGraph workflows for document retrieval, validation, and reasoning, streamlining query resolution across enterprise knowledge bases.
- Architected agent-based orchestration pipelines integrating LLMs, external tools, and vector databases to automate document processing and decision-support workflows that previously required manual review.
- Built document preprocessing and retrieval pipelines covering **PDF extraction, chunking, embeddings, metadata filtering, and semantic search**, improving retrieval quality for document-based QA.
- Implemented **RAG evaluation and testing** using relevance, faithfulness, and retrieval-quality metrics to identify failures in chunking, retrieval, prompting, and generation.
- Implemented **LLM observability and tracing** to monitor **latency, token usage, retrieval results, prompts, model responses, and pipeline failures**, supporting debugging and production optimization.
- Partnered with product, backend, and engineering teams to deploy AI solutions into enterprise production workflows.
- Delivered proof-of-concepts (PoCs) in predictive analytics, computer vision, and time-series forecasting, several of which advanced toward production development.

PROJECTS

Compliance Document RAG Assistant

Tech Stack: Python, LangChain, LangGraph, FAISS, Azure OpenAI, Vector Embeddings

- Developed a RAG-based chatbot for answering regulatory and compliance queries using relevant document context.
- Built the ingestion pipeline covering **PDF/text/table extraction, chunking, embeddings, and vector indexing**.
- Converted complex regulatory tables into structured text before indexing so approval authorities, financial limits, and procedures could be retrieved accurately.
- Implemented hybrid retrieval combining keyword and vector search with metadata filtering and reranking to improve retrieval precision for regulatory queries.
- Designed an **evaluation framework** using retrieval relevance, context quality, and answer faithfulness to identify RAG failure cases and improve response quality.
- Used LangGraph to orchestrate retrieval and response workflows and **traced pipeline execution to monitor latency, retrieval results, and LLM outputs**.

AI-Based Document Intelligence Platform

Tech Stack: Python, AWS Bedrock, LangChain, LangGraph, FastAPI

- Developed an AI-powered document intelligence pipeline using a multimodal Bedrock model to understand uploaded documents.
- Implemented document classification and mapping to predefined document checklists, identifying which required document each upload represents.

- Extracted relevant information from classified documents and converted the results into structured data for downstream processing.
- Combined LLM-based extraction with rule-based validation to improve consistency and support business workflows.
- Exposed document-processing capabilities through FastAPI for integration with backend applications.

PDF Table Detection & Extraction

Tech Stack: Python, YOLO, OpenCV, Tesseract OCR, pdfplumber, Pandas

- Developed a PDF table extraction system for processing documents containing text, images, and different table layouts.
- Trained a YOLO-based model to detect bordered, borderless, and different types of tables within PDF pages.
- Applied different extraction techniques based on the detected table structure to handle both standard and irregular tables.
- Used PDF parsing and OCR techniques to extract table content and reconstruct it into structured rows and columns.
- Converted extracted table data into Excel and structured formats for downstream processing.
- Handled challenging PDFs with borderless tables, irregular layouts, images, and vertically oriented text.

Fire, Smoke & PPE Detection System (PoC)

Tech Stack: Python, YOLO, OpenCV, Raspberry Pi

- Developed a computer-vision PoC to detect fire, smoke, gloves, and PPE from images/video.
- Prepared and annotated training data and trained a YOLO-based object detection model.
- Evaluated the trained model on test images and video to verify detection performance.
- Deployed the PoC on Raspberry Pi for edge-based inference.

SKILLS

Programming: Python

Generative AI & LLMs: Agentic AI, LangChain, LangGraph, LLMs, Prompt Engineering, RAG

RAG & Evaluation: Embeddings, Hybrid Search, LLM Observability, RAG Evaluation, Reranking, Vector Search

Vector Databases: Azure Cosmos DB, Chroma, FAISS, pgvector

Document AI & OCR: Document Indexing, OCR, PDF Extraction, Table Parsing

Computer Vision: MediaPipe, OpenCV, YOLO

Machine Learning: Classification, PyTorch, Regression, Scikit-learn, Time-Series Forecasting

Backend & Cloud: AWS Bedrock, Azure OpenAI, Docker, FastAPI, RESTful APIs

Data & Tools: Git, GitHub, NumPy, Pandas, Streamlit

EDUCATION

B.Tech in Computer Science and Engineering

Amal Jyothi College of Engineering | A.P.J Abdul Kalam Technological University

08/2018 - 06/2022